

Supplementary Material

ESM_2: Sensitivity Analysis Tables (Full 2006-2024 Sample)

Article:

Smart City Construction, Sustainable Urban Transition, and Carbon Emissions: Nonlinear Evidence from Chinese Cities

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Descriptive statistics: full sample, 2006-2024

	count	mean	p50	sd	min
CE	5396	4.612	4.654	1.253	0.000
SCCD	5396	0.318	0.304	0.085	0.121
Spatial coupling coordination degree squared	5396	0.108	0.092	0.066	0.015
OPEN	5381	0.193	0.080	0.337	-0.720
UR	5396	54.498	52.606	15.419	6.491
URG	5381	2.437	2.342	0.549	1.207
GI	5396	0.186	0.162	0.098	0.043
ER	5396	0.008	0.007	0.003	0.000
DEI	5396	0.060	0.047	0.076	0.000
POLY	5396	0.346	0.374	0.189	0.000
OIU	5396	2.288	2.283	0.144	1.163
GTI	5396	4.047	4.038	1.795	0.000
MCCD	5396	0.318	0.304	0.085	0.121
NTL	5396	1.421	1.434	1.128	-2.322
FDI	5365	2.244	2.282	0.277	-1.693
RD	5396	0.015	0.010	0.016	0.000
EI	5396	-0.161	-0.083	0.547	-5.026
ISL	5396	0.416	0.412	0.099	0.038
ISU	5396	1.034	0.911	0.563	0.094
IV	5112	0.316	0.302	0.085	0.121
c_SCCD	5396	0.005	-0.009	0.085	-0.192
is_fitted	5396	0.158	0.000	0.365	0.000

Baseline regression: full sample, 2006-2024

	(1) CE	(2) CE	(3) CE	(4) CE	(5) CE
SCCD	13.468*** (1.546)	12.344*** (1.521)	9.123*** (1.374)	8.871*** (1.408)	8.378*** (1.388)
Spatial coupling coordination degree squared	-11.472*** (1.814)	-10.369*** (1.816)	-8.283*** (1.622)	-7.882*** (1.658)	-7.725*** (1.665)
OPEN		0.486*** (0.107)	0.204** (0.084)	0.223*** (0.085)	0.259*** (0.084)
UR			0.032*** (0.004)	0.030*** (0.004)	0.028*** (0.004)
URG				-0.135* (0.073)	-0.112 (0.070)
GI					-2.064*** (0.626)
Constant	1.057*** (0.311)	1.171*** (0.299)	0.438 (0.284)	0.944** (0.377)	1.400*** (0.412)
N	5396	5381	5381	5366	5366
Within R-squared	0.568	0.579	0.629	0.631	0.641

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness and IV-2SLS: full sample, 2006-2024

	(1) w_CE	(2) NTL	(3) CE	(4) CE
w_SCCD	9.888*** (1.414)			
w_SCCD2	-9.886*** (1.698)			
w_OPEN	0.355*** (0.111)			
w_UR	0.028*** (0.004)			
w_URG	-0.076 (0.073)			
w_GI	-2.007*** (0.619)			
SCCD		3.536*** (0.757)		18.658*** (2.694)
SCCD2		-2.515*** (0.789)		-14.618*** (2.711)
OPEN		0.252*** (0.088)	0.259*** (0.084)	0.153* (0.083)
UR		0.020*** (0.003)	0.028*** (0.004)	0.019*** (0.004)
URG		-0.053 (0.063)	-0.112 (0.070)	-0.108 (0.066)
GI		-1.532*** (0.333)	-2.064*** (0.626)	-1.541** (0.664)
MCCD			8.378*** (1.388)	
MCCD2			-7.725*** (1.665)	
Constant	1.030**	-0.412	1.400***	-0.439

	(0.414)	(0.299)	(0.412)	(0.564)
N	5366	5366	5366	5084
Within R-squared	0.647	0.573	0.641	0.602

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

First-stage regressions: full sample, 2006-2024		
	(1)	(2)
	SCCD	SCCD2
IV	0.344*** (0.064)	-0.035 (0.051)
Squared IV based on lagged SCCD	-0.071 (0.079)	0.382*** (0.070)
OPEN	0.012* (0.006)	0.006 (0.005)
UR	0.001*** (0.000)	0.001*** (0.000)
URG	0.009** (0.003)	0.010*** (0.003)
GI	-0.101*** (0.021)	-0.073*** (0.015)
Constant	0.112*** (0.018)	-0.004 (0.013)
N	5084	5084
Within R-squared	0.652	0.586

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Regional heterogeneity: full sample, 2006-2024

	(1) Eastern China	(2) Central China	(3) Western China
SCCD	7.589*** (2.677)	12.171*** (2.747)	7.221*** (1.831)
SCCD2	-6.922*** (2.582)	-11.725*** (3.489)	-7.151*** (2.180)
OPEN	0.219** (0.093)	0.352 (0.274)	0.266 (0.289)
UR	0.030*** (0.006)	0.024*** (0.007)	0.028*** (0.007)
URG	-0.036 (0.141)	-0.037 (0.138)	-0.205* (0.105)
GI	-0.730 (1.130)	-2.461*** (0.754)	-2.017** (0.999)
Constant	1.278* (0.740)	0.523 (0.763)	1.946*** (0.675)
N	1828	1885	1653
Within R-squared	0.720	0.629	0.621

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Mediation analysis: full sample, 2006-2024

	(1) OIU	(2) CE with OIU	(3) GTI	(4) CE with GTI
SCCD	0.626*** (0.163)	7.445*** (1.297)	10.481*** (1.386)	4.971*** (1.156)
SCCD2	-0.371* (0.213)	-7.172*** (1.486)	-5.438*** (1.416)	-5.957*** (1.418)
OIU		1.491*** (0.337)		
GTI				0.325*** (0.030)
OPEN	0.004 (0.013)	0.253*** (0.079)	0.241** (0.094)	0.180** (0.075)
UR	0.004*** (0.000)	0.022*** (0.004)	0.022*** (0.005)	0.021*** (0.004)
URG	0.028*** (0.006)	-0.153** (0.069)	0.016 (0.067)	-0.117* (0.065)
GI	-0.088* (0.051)	-1.933*** (0.595)	-1.908*** (0.396)	-1.444** (0.649)
Constant	1.811*** (0.043)	-1.300* (0.738)	-0.817* (0.423)	1.666*** (0.355)
N	5366	5366	5366	5366
Within R-squared	0.614	0.651	0.812	0.683

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Moderation analysis: full sample, 2006-2024

	(1) ER	(2) DEI	(3) POLY
SCCD	9.569*** (1.995)	8.829*** (1.289)	7.770*** (2.188)
SCCD2	-9.753*** (2.463)	-7.714*** (1.473)	-8.223*** (2.661)
ER	33.864 (45.908)		
SCCD # ER	-158.794 (230.885)		
SCCD2 # ER	260.988 (277.816)		
OPEN	0.258*** (0.085)	0.235*** (0.087)	0.260*** (0.084)
UR	0.028*** (0.004)	0.028*** (0.004)	0.027*** (0.004)
URG	-0.108 (0.070)	-0.100 (0.070)	-0.100 (0.070)
GI	-2.068*** (0.608)	-2.119*** (0.622)	-2.049*** (0.606)
DEI		4.601** (2.080)	
SCCD # DEI		-22.377*** (7.346)	
SCCD2 # DEI		21.644*** (6.145)	
POLY			-1.179 (0.970)
SCCD # POLY			3.213 (4.848)
SCCD2 # POLY			-0.186

			(5.505)
Constant	1.157** (0.498)	1.267*** (0.404)	1.699*** (0.514)
N	5366	5366	5366
Within R-squared	0.642	0.643	0.644
Standard errors in parentheses			
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$			